



Algoritmi di classificazione a supporto di diagnosi mediche





Dataset

Dimensionalità: 373x15

Variabili: Id soggetto, Id risonanza magnetica, gruppo, visita, ritardo mri, età, n° anni istruzione, sesso, mano dominante, status socioeconomico, punteggio MMSE, Clinical Dementia Rating, Volume intracranico stimato, Volume cerebrale normalizzato, fattore di scala

NA: 21

Domanda

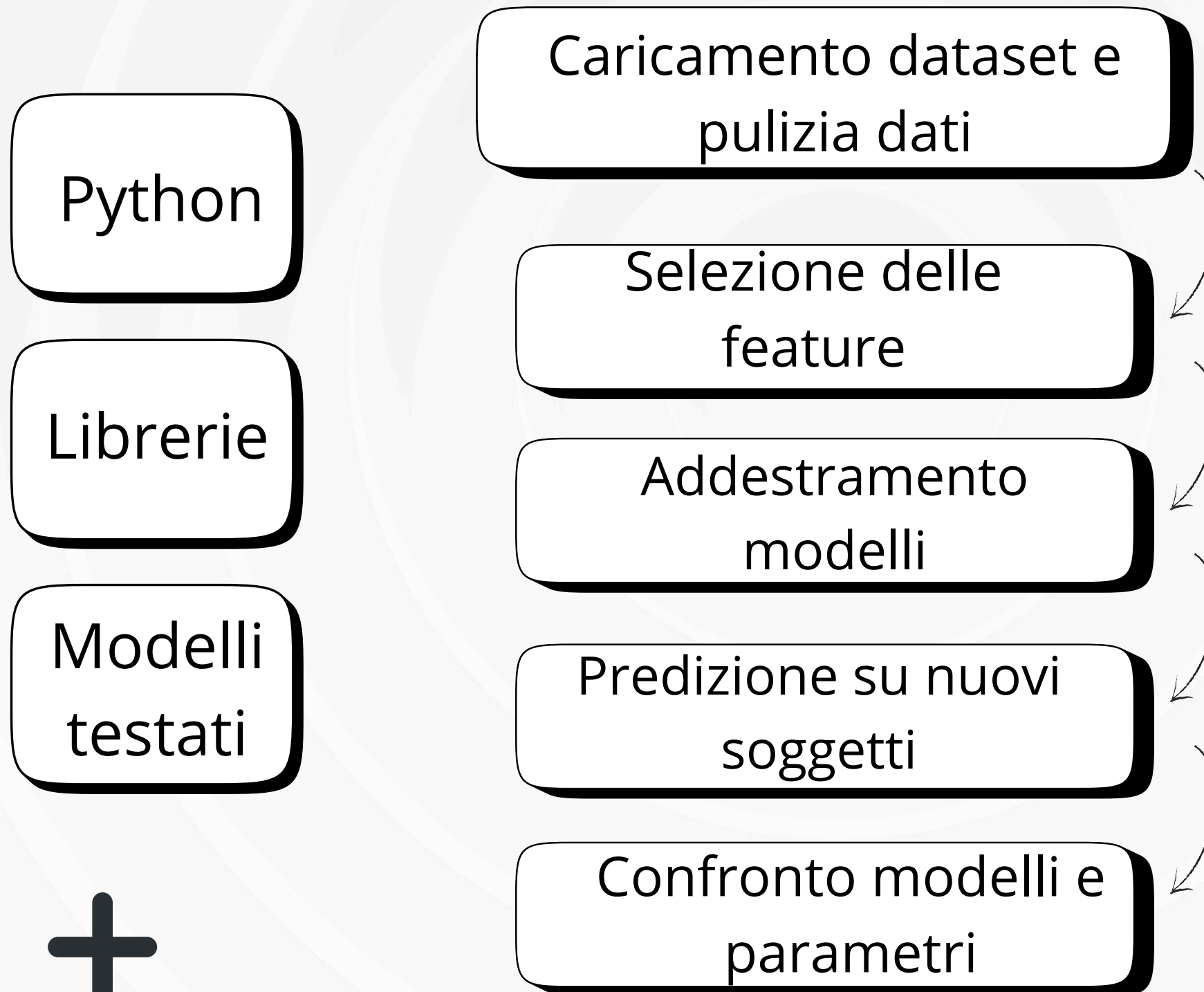
È possibile prevedere l'insorgenza o la progressione della demenza (es. Alzheimer) in base ai dati clinici e MRI longitudinali dei pazienti?

Obiettivo

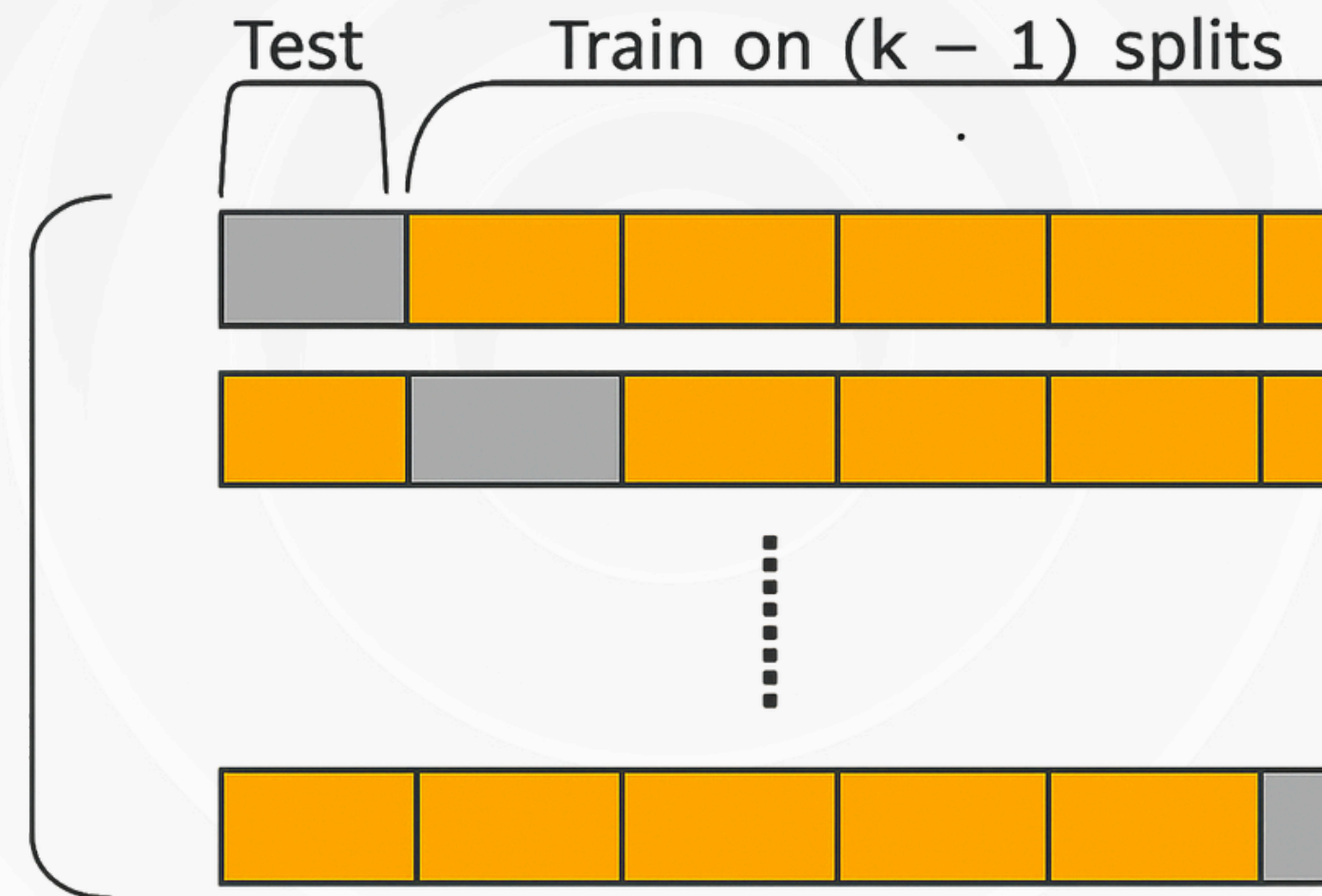
Strumenti di machine learning come strumenti di
sostegno nella diagnosi in ambito medico

Metodologia

strumenti e procedimento



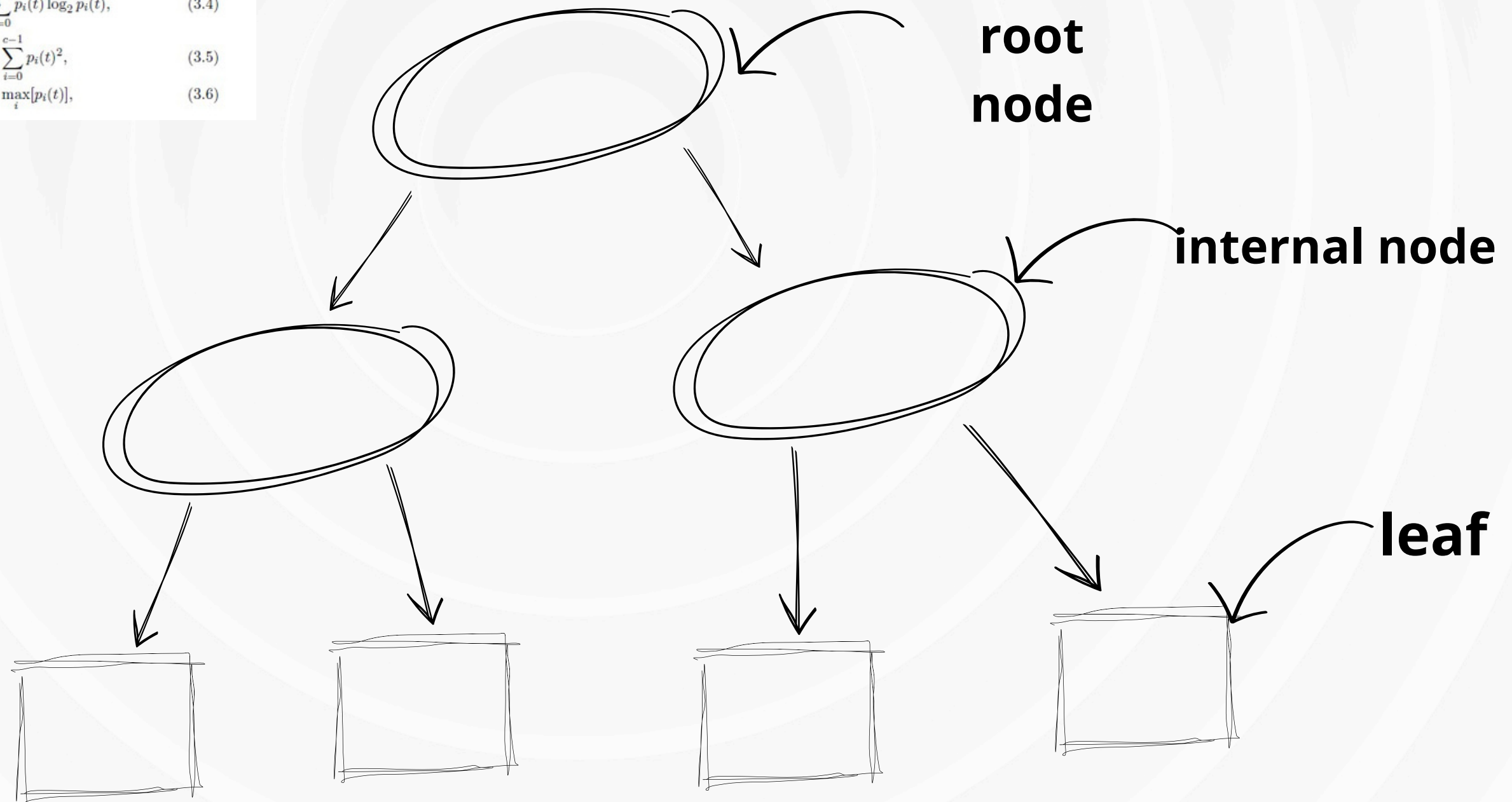
Leave One Out



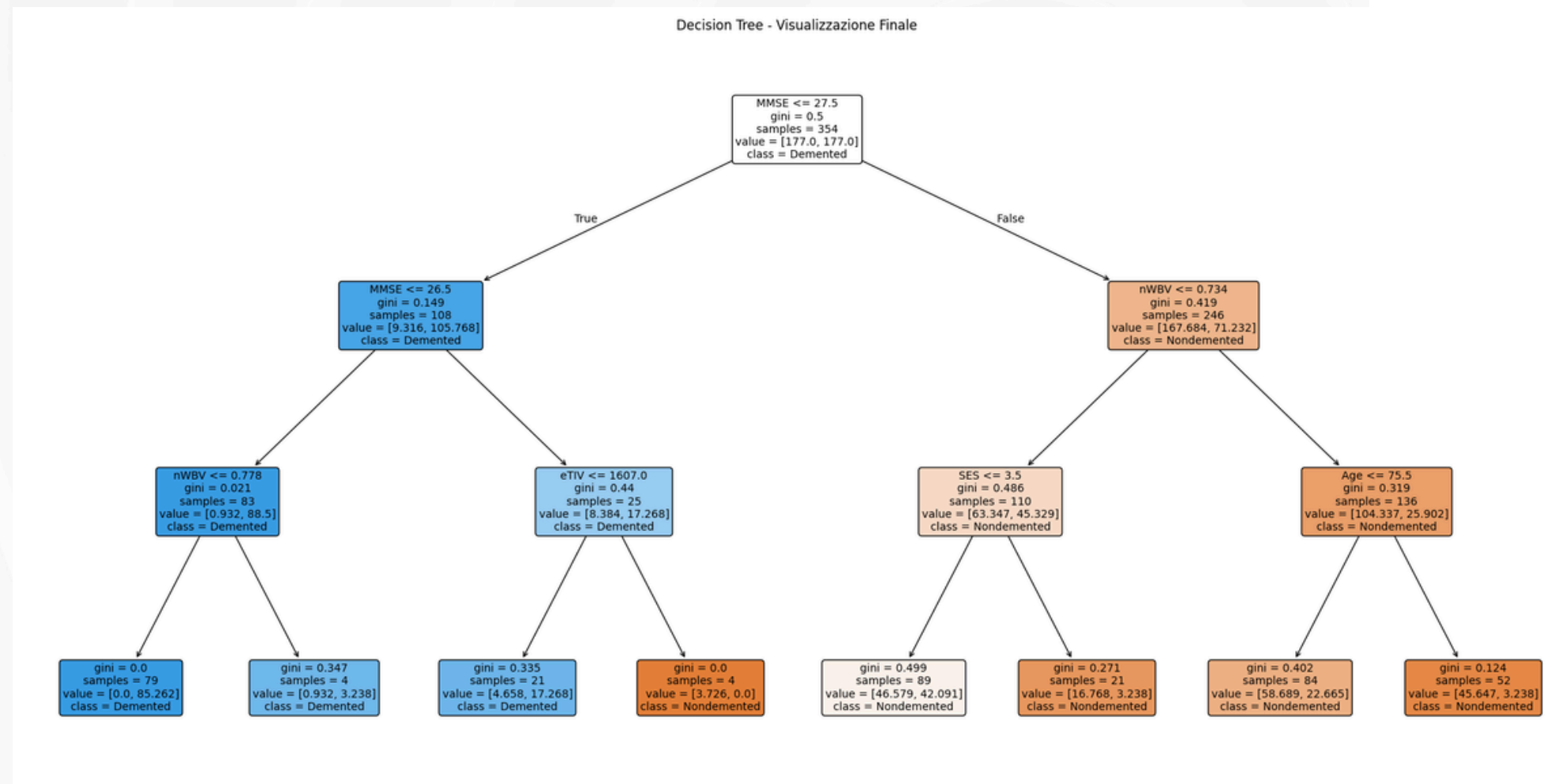
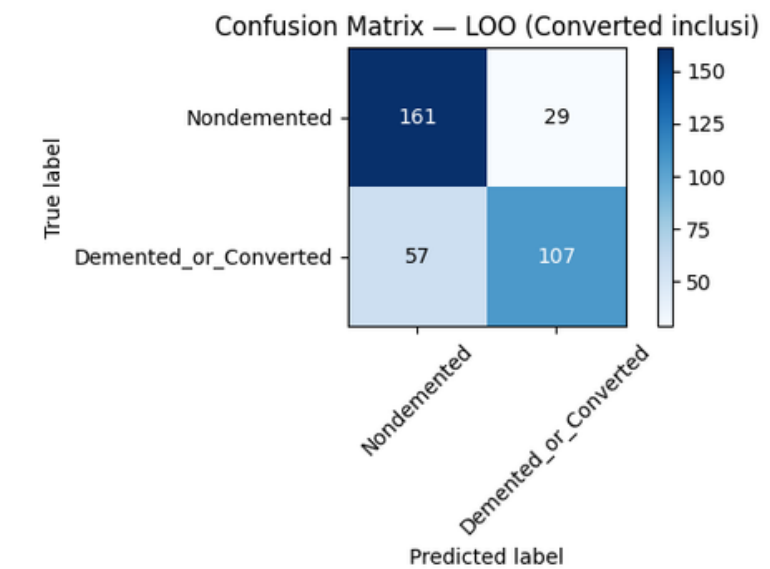
Decision Tree

Come funziona?

$$\begin{aligned}\text{Entropy} &= -\sum_{i=0}^{c-1} p_i(t) \log_2 p_i(t), & (3.4) \\ \text{Gini index} &= 1 - \sum_{i=0}^{c-1} p_i(t)^2, & (3.5) \\ \text{Classification error} &= 1 - \max_i [p_i(t)], & (3.6)\end{aligned}$$

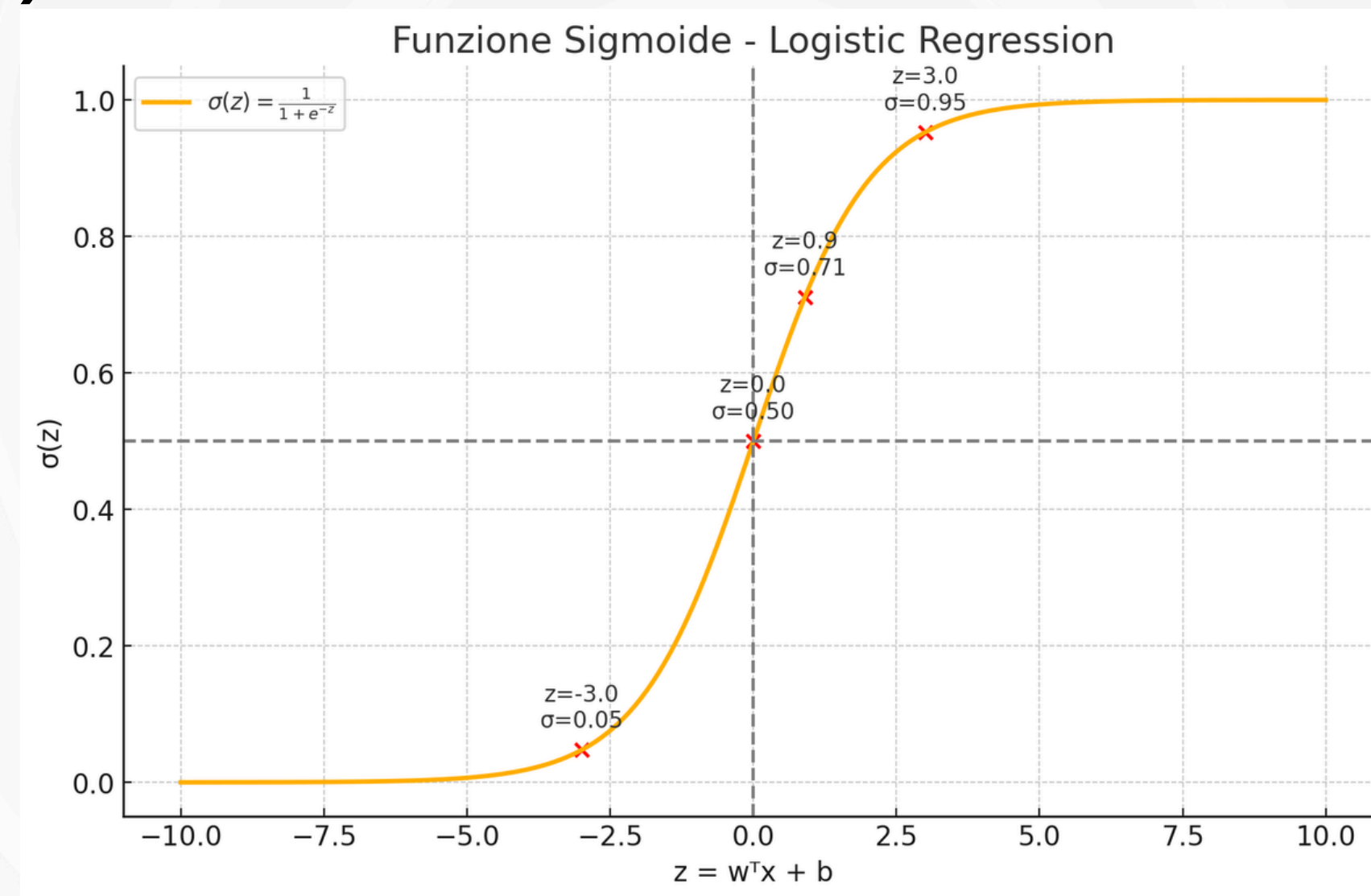


Output Decision Tree

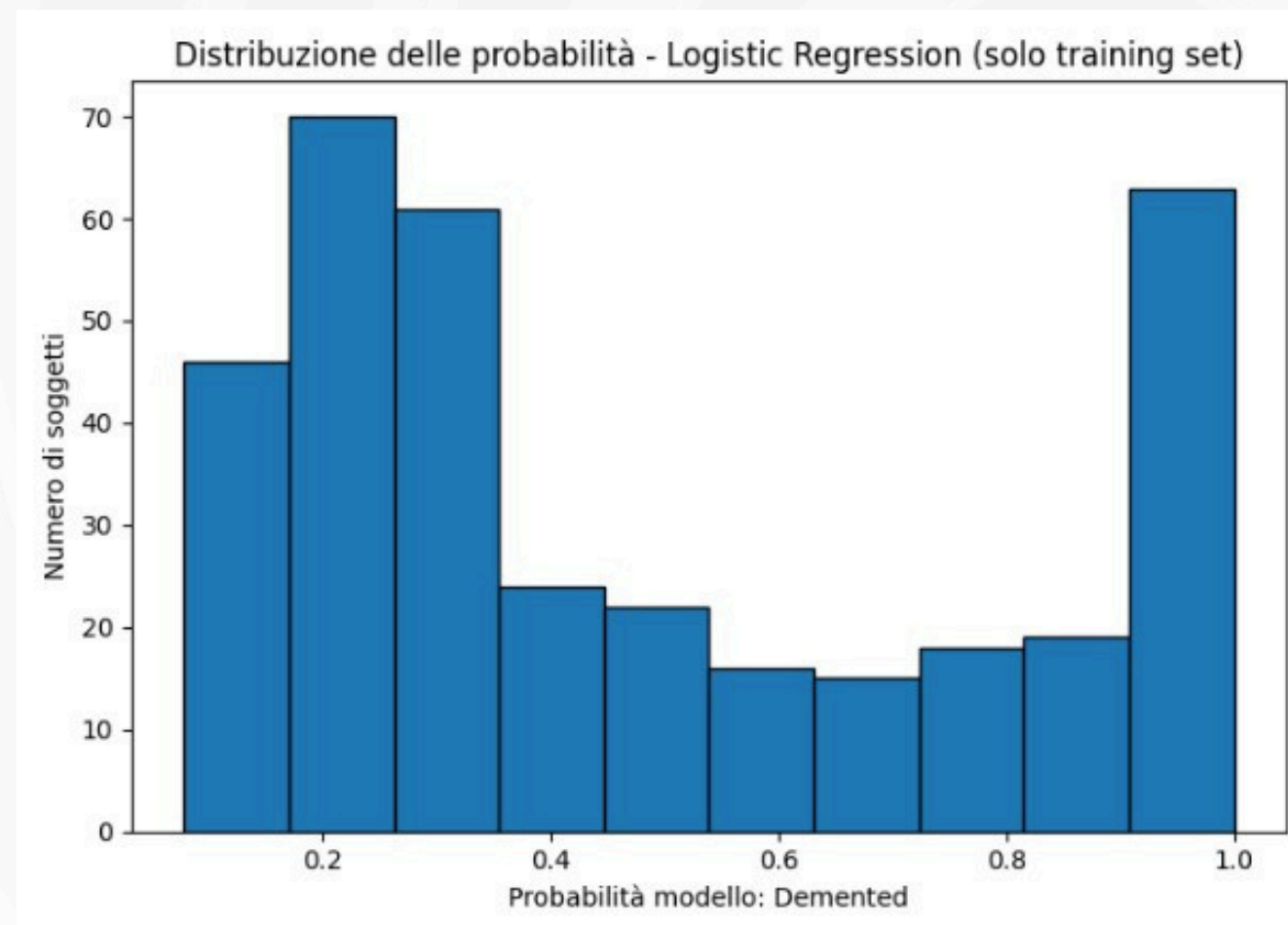


Logistic Regression

$P(X|Y)$ vs $P(Y|X)$



Output Logistic Regression



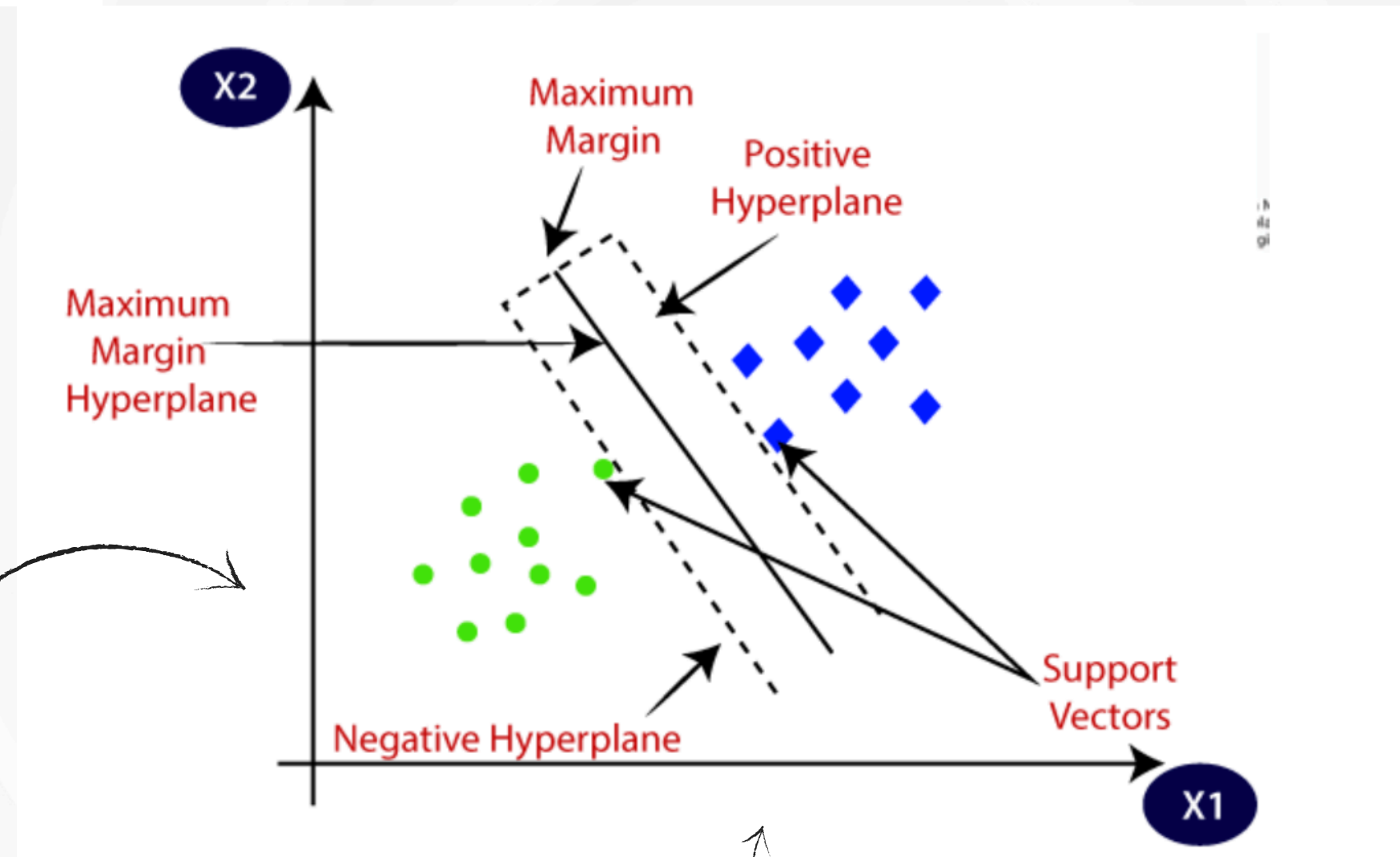
Confusion Matrix — Logistic Regression (LOO)

True	Predicted	
	Nondemented	Demented
Nondemented	162	28
Demented	52	112

Support Vector Machine

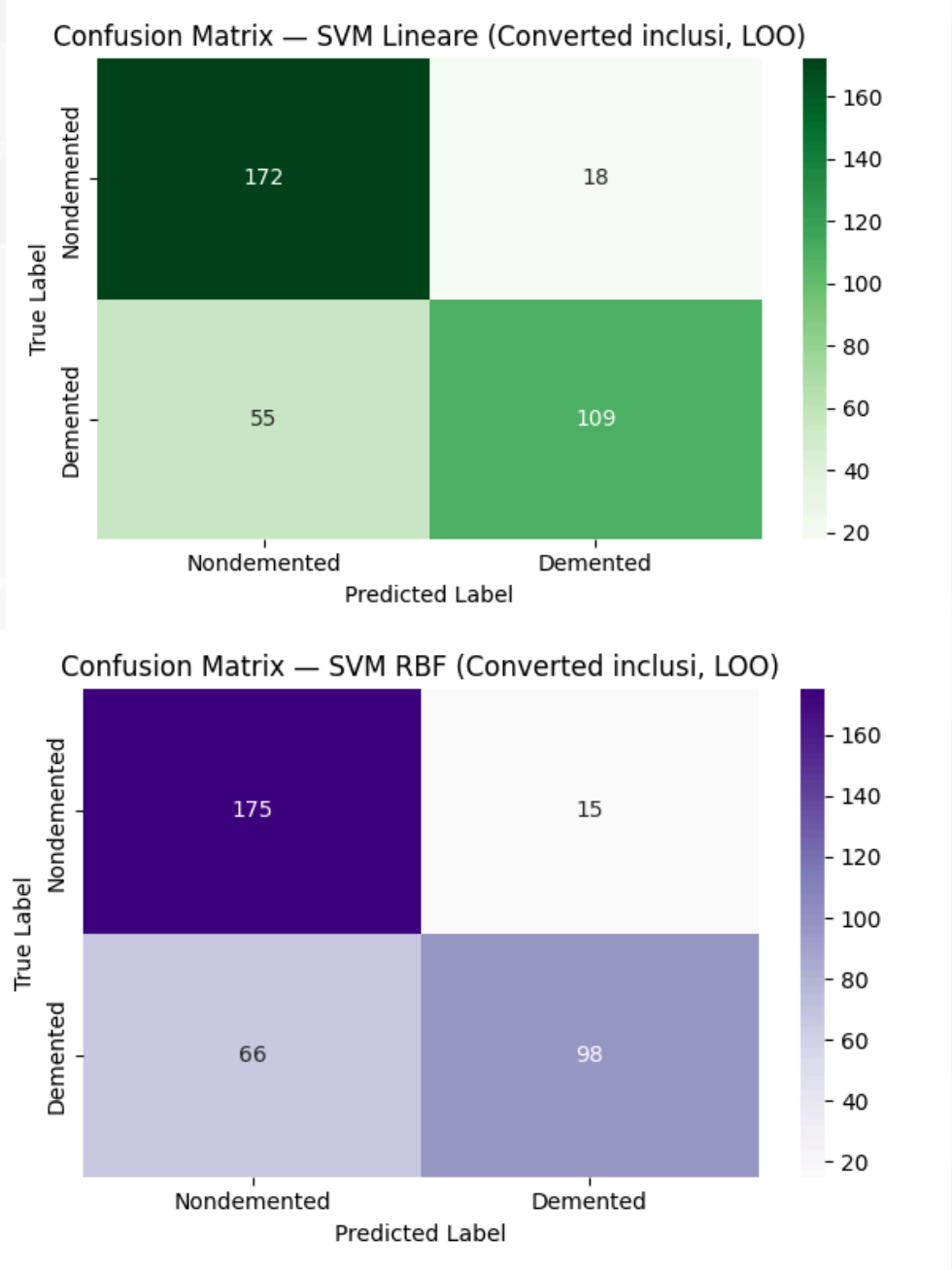
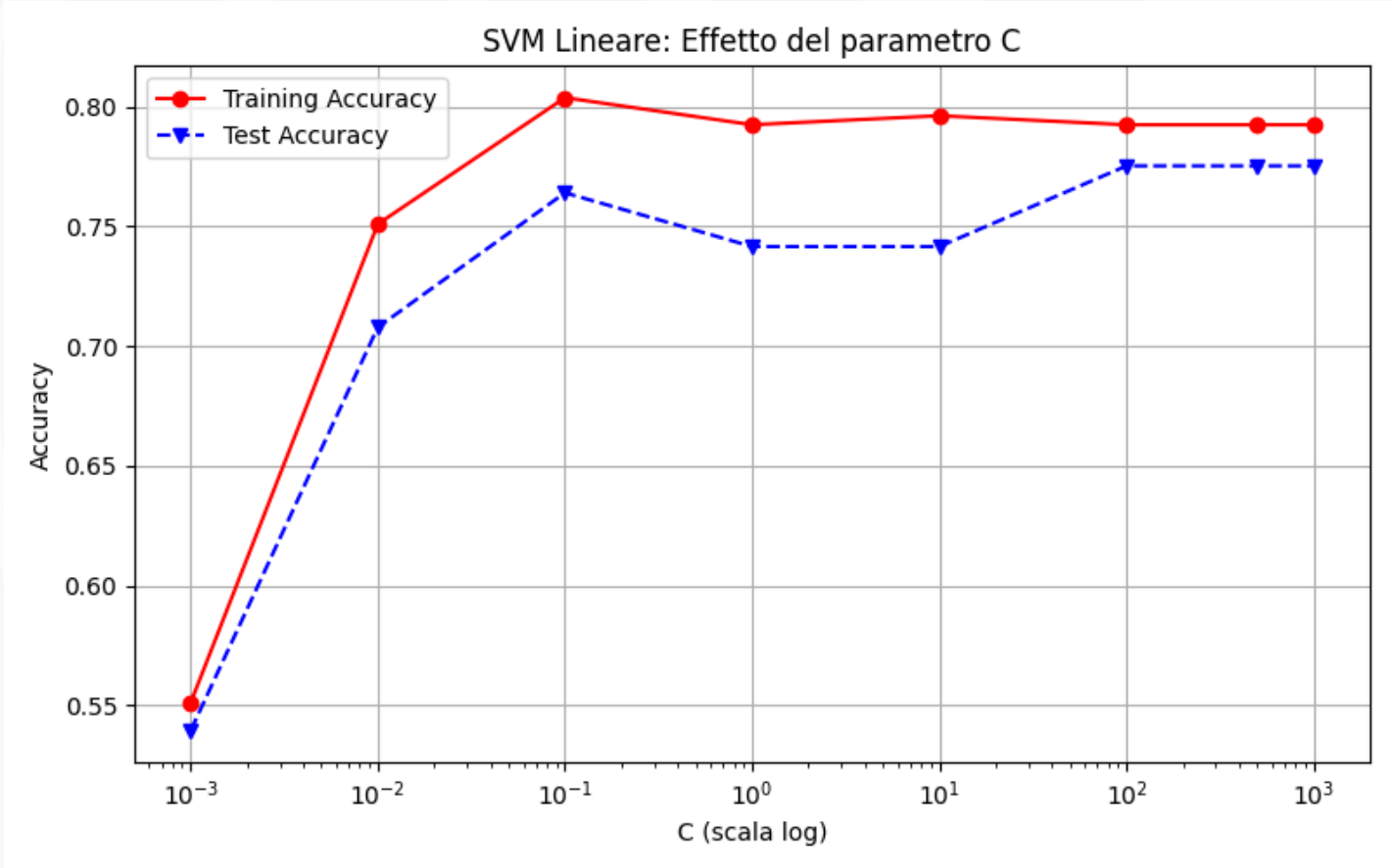
$$w^T x + b = 0,$$

nWBV



MMSE

Output Vector Machine



Discussione

Decision Tree

```
clf = tree.DecisionTreeClassifier(  
    criterion='gini',  
    max_depth=3,  
    min_samples_split=2,  
    min_samples_leaf=1,  
    class_weight = 'balanced',  
    random_state=42  
)
```

Classification Report:

	precision	recall	f1-score	support
Nondemented	0.71	0.87	0.78	190
Demented_or_Converted	0.80	0.59	0.68	164
accuracy			0.74	354
macro avg	0.75	0.73	0.73	354
weighted avg	0.75	0.74	0.73	354

SVM

```
svm_linear = Pipeline([  
    ('scaler', StandardScaler()),  
    ('svc', SVC(kernel='linear', C=1,  
    class_weight='balanced',  
    random_state=42))  
])
```

=== CLASSIFICATION REPORT (SVM LINEARE, L00CV) ===

	precision	recall	f1-score	support
Nondemented	0.76	0.91	0.82	190
Demented/Converted	0.86	0.66	0.75	164
accuracy			0.79	354
macro avg	0.81	0.78	0.79	354
weighted avg	0.80	0.79	0.79	354

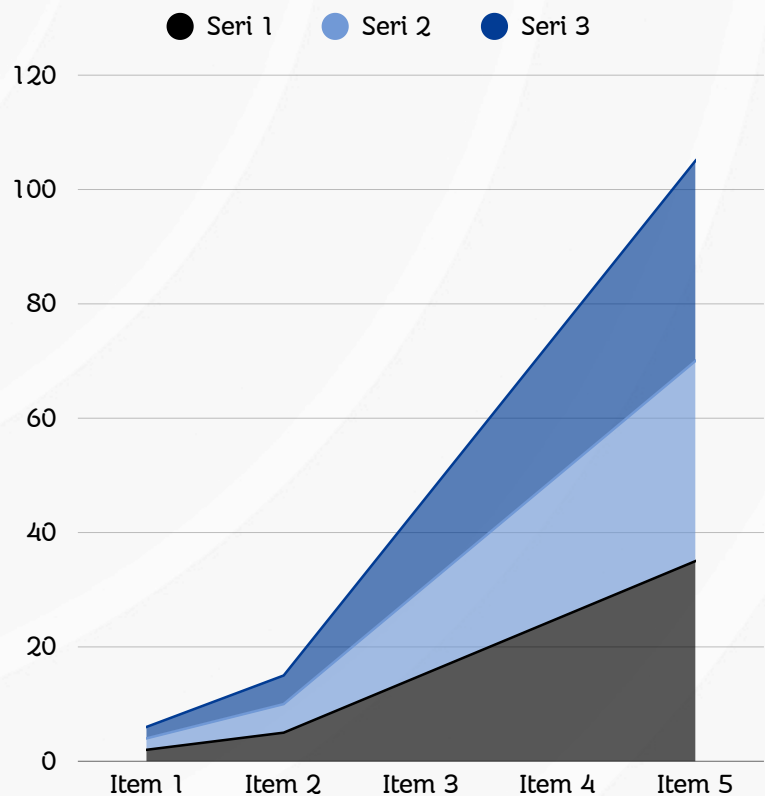
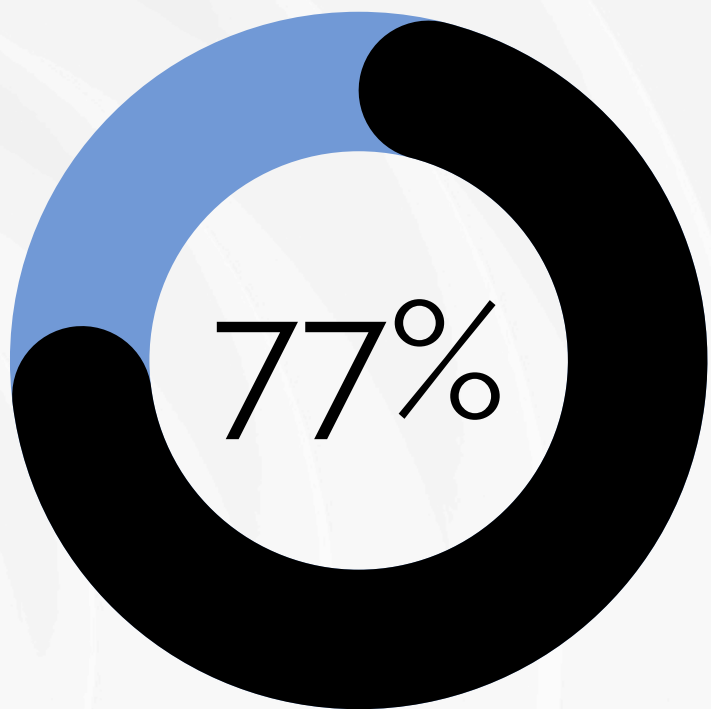
Accuracy complessiva (SVM Lineare, L00): 0.7938

Regressione logistica

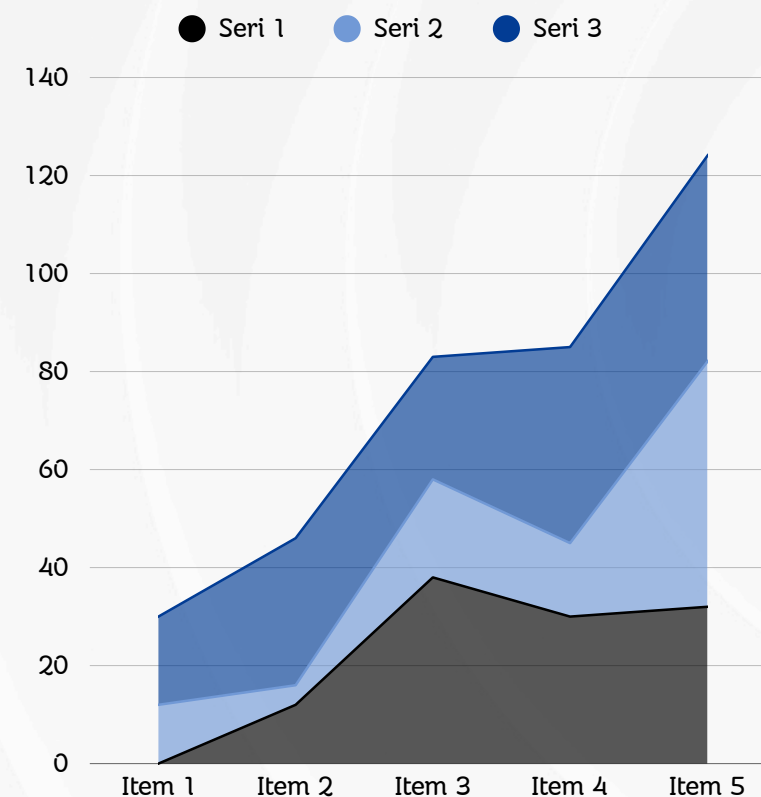
```
log_reg =  
LogisticRegression(max_iter=100  
0, class_weight='balanced')
```

=== CLASSIFICATION REPORT (L00) ===

	precision	recall	f1-score	support
Nondemented	0.76	0.85	0.80	190
Demented/Converted	0.80	0.68	0.74	164
accuracy			0.77	354
macro avg	0.78	0.77	0.77	354
weighted avg	0.78	0.77	0.77	354



Manipolazione iperparametri



Iperparametri testati

- **Decision Tree:** entropy vs gini
- max depth
- **Logistic Regression:** max iter -
class_weight
- **Support Vector Machine:**
kernel, C, class weight

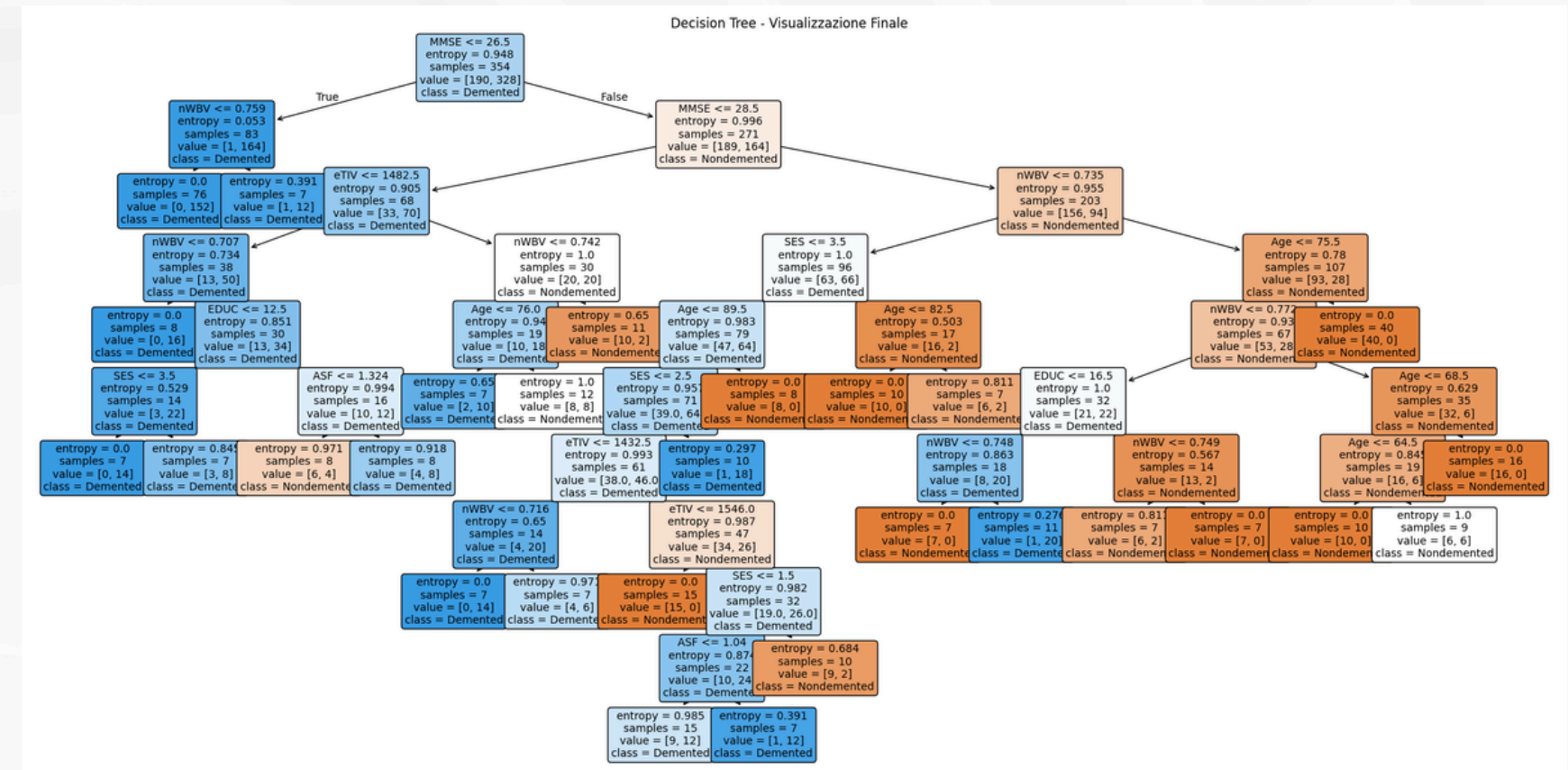
Obiettivo: Alzare il punteggio di Recall sulla demenza

+ Risultati Decision Tree

post modifiche iperparametri

criterion='entropy',
max_depth=7,
min_samples_split=2,
min_samples_leaf=3,
class_weight = {0: 1, 1: 2},
random_state=42

Classification Report:				
	precision	recall	f1-score	support
Nondemented	0.82	0.80	0.81	190
Demented_or_Converted	0.77	0.79	0.78	164
accuracy			0.80	354
macro avg	0.80	0.80	0.80	354
weighted avg	0.80	0.80	0.80	354



Confusion Matrix — LOO (Converted inclusi)

True label	Predicted label	
	Nondemented	Demented_or_Converted
Nondemented	142	48
Demented_or_Converted	31	133

Discussione

ora sorge la problematica del fit dei dati

```
=== CONFRONTO ACCURACY - OVERFITTING CHECK ===  
Training Accuracy: 0.9096  
LOO Accuracy      : 0.7966  
Overfitting Gap    : 0.1130
```

GAP

= 0.00-0.05



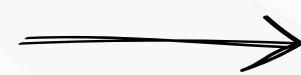
Generalizzazione buona - modello stabile

0.05 - 0.15



Potenziiale overfitting, valutare potature o riduzione profondità

> 0.15



Overfitting marcato, iperparametri da rivedere

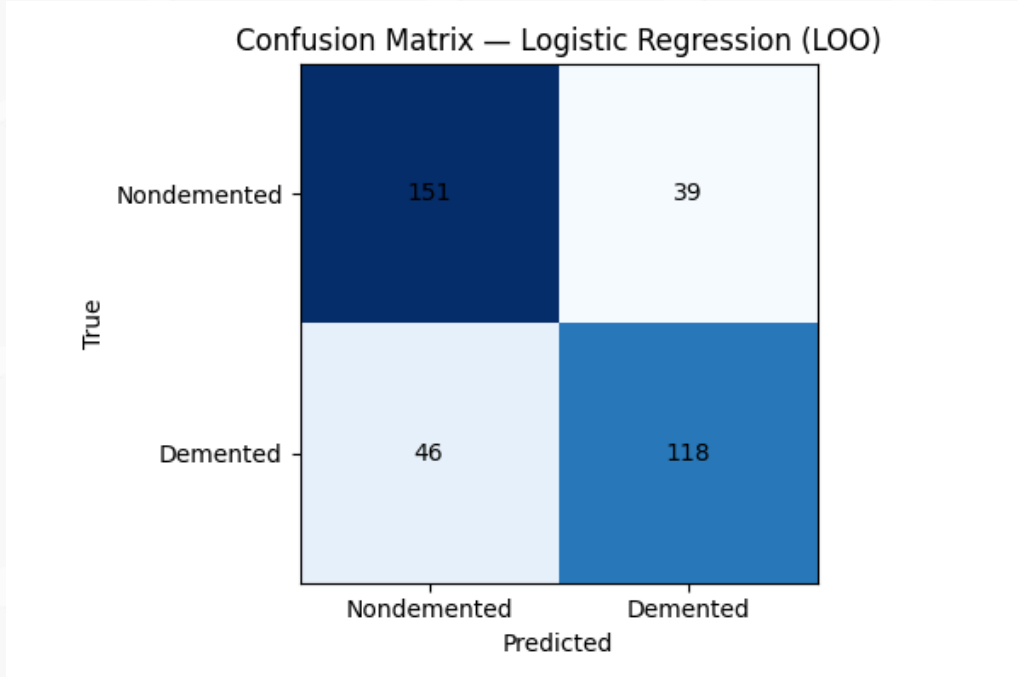
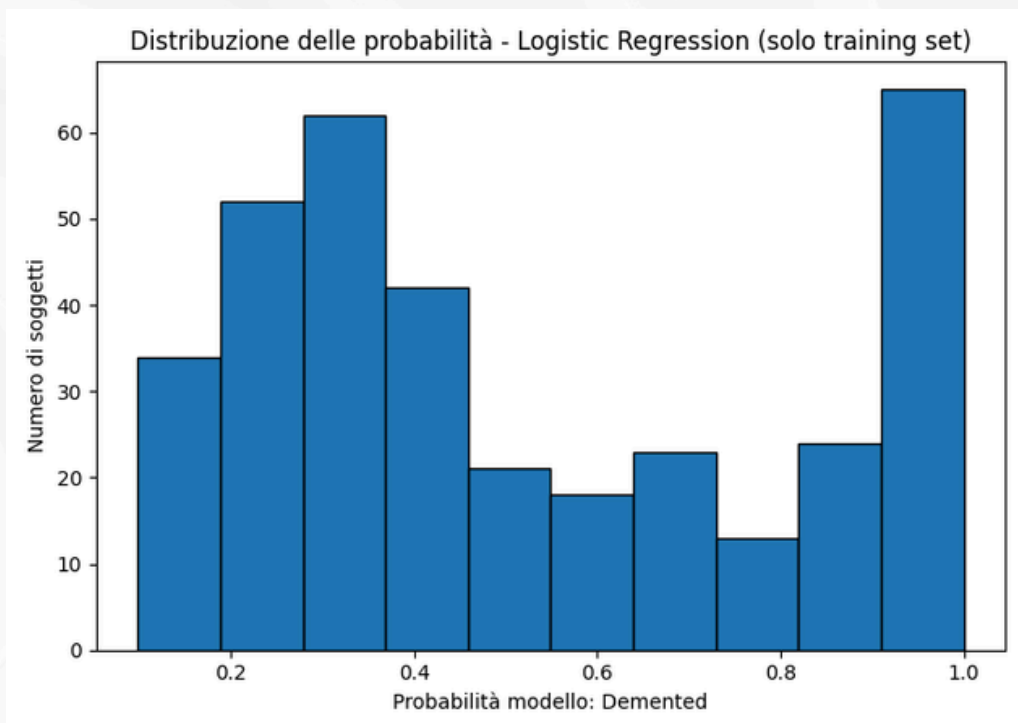


Risultati Logistic Regression

post modifica iperparametri

```
=== CLASSIFICATION REPORT (LOO) ===
```

	precision	recall	f1-score	support
Nondemented	0.77	0.79	0.78	190
Demented/Converted	0.75	0.72	0.74	164
accuracy			0.76	354
macro avg	0.76	0.76	0.76	354
weighted avg	0.76	0.76	0.76	354





Discussione

post modifiche iperparametri

```
=== CONFRONTO ACCURACY ===  
Training Accuracy: 0.7627  
LOOCV Accuracy   : 0.7599  
Overfitting Gap   : 0.0028
```

In questo caso ho
aumentato il peso di **1.5** e le
iterazioni a **1500**



Risultati SVM

post modifica iperparametri

Il Support Vector Machine presentava livelli di accuratezza analoghi ai modelli precedenti con rischio di overfitting più alto



Confronto modelli

